The Adverse Effect Wage Rate and H-2A Program Use

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December 31, 1979

Abstract

The H-2A program permits U.S. agricultural employers to hire temporary foreign workers but requires them to pay at least the Adverse Effect Wage Rate (AEWR). We construct a county-year panel for 2008–2022 that combines geocoded H-2A certifications, AEWRs, local wage measures, agricultural employment, and county characteristics. Our difference-in-differences design compares counties facing different lagged AEWR premiums relative to the 25th percentile of the wage distribution in their commuting zone before and after 2011. A larger premium is associated with lower H-2A use after 2011: the controlled estimate is -0.0061 percentage points of 2011 farm employment per additional 2012 dollar of the lagged premium. Event-study estimates are near zero before 2011 and become increasingly negative after 2014. We also report exploratory evidence on agricultural prices and labor expenditure shares, which should not be interpreted as a complete account of downstream adjustment.

1 Introduction

The H-2A program is the principal uncapped temporary foreign-worker program for U.S. agriculture. Its use has expanded sharply over the past two decades, as illustrated in Figure 1. At the same time, the program imposes a distinctive wage rule: employers must pay the highest applicable wage floor, which commonly is the AEWR. Because employers also incur recruitment, housing, transportation, and administrative costs, the AEWR can affect both the decision to participate in H-2A and the scale of certified employment.

This paper asks whether a higher AEWR relative to local outside wages reduces H-2A program use. The empirical analysis uses county-year observations. The outcome is certified H-2A workers with a contract start date in the calendar year, normalized by county farm employment in 2011. The key regressor is the one-year-lagged, real AEWR minus the 25th percentile local wage in the county's commuting zone. This definition measures the wage premium relevant to a farm's decision more directly than a comparison with the statutory minimum wage.

Our baseline specification is a continuous-treatment difference-in-differences model with county and year fixed effects. It compares changes after 2011 in counties with relatively high and low AEWR wage premiums. In the controlled model, a one-dollar larger lagged premium is associated with a 0.0061 percentage-point lower H-2A certification rate (Table 2). The estimate is similar when commuting zones that span AEWR-region boundaries are excluded. The event study provides no clear evidence of differential pre-trends and shows increasingly negative estimates in later post-2011 years.

The results describe the reduced-form relationship generated by this design. They should not be read as a simple elasticity of H-2A demand, nor as evidence that every component of a change in the AEWR is exogenous. We therefore also develop a separate sampling-composition instrument for AEWR changes; that work is currently limited to first-stage diagnostics and is not used for the main causal estimates.

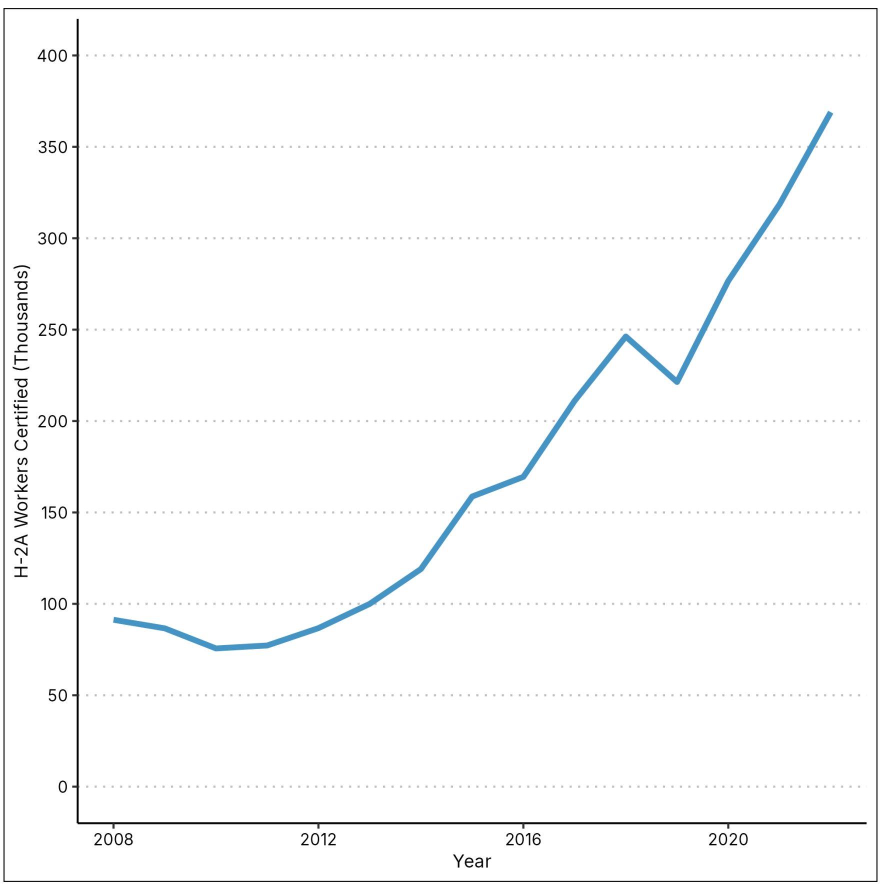

Figure 1: Annual H-2A workers certified.

2 H-2A agricultural guest worker program

2.1 Background of the program

The Bracero program, which was the forerunner to the H-2A program, was created in 1942 to fill labor shortages due to the war. The program allowed migrant workers, mostly Mexicans, to temporarily work in the U.S. agricultural industry. The program was terminated in 1964, and was succeeded by the H-2A guest worker program which was created as part of the Immigration Reform and Control Act of 1986. The act also created the H-1B and H-2B guest worker programs. The H-2A program is unique among major U.S. temporary worker programs in that it does not have a numerical cap; its use has consequently grown substantially since the mid-2000s. The H-2A program requires employers to complete an expensive multi-step certification process, involving multiple state and federal agencies. The positions filled via the H-2A program has to be full-time, in the agricultural sector, and seasonal in nature. Note that this traditionally excludes many jobs involving animal husbandry due to their year-round nature.

2.2 H-2A hiring process

The U.S. Department of Agriculture (USDA) and Department of Labor (DOL) provide information to prospective employers who wish to use the program (United States Department of Agriculture 2025a; United States Department of Labor 2025b). We summarize the standard (non-expedited) application process below. This process begins about 3 months before the actual start of work.

1. The farmer first submits a job order to their state's State Workforce Agency (SWA), which notifies the SWA of the nature of the future H-2A application and requests an inspection of the housing provided by the employer. The SWA, once it accepts the job order, begins domestic recruitment for the job order.

2. The farmer simultaneously applies for labor certification with the Department of Labor's Office of Foreign Labor Certification (OFLC)

3. Once the labor certification application is accepted by the OFLC, the farmer has to begin domestic recruitment for the H-2A positions. The OFLC provides instructions to the farmer for this process. Domestic recruitment may involve:

    - Posting newspaper ads for the positions in a local newspaper that serves the area around the worksite

- Contacting previously hired domestic workers who were not previously fired for cause or abandoned the job, offering them a position

- Advertising the position in a maximum of three other states in the region or where they expect to hire (domestic) workers

   The farmer has to interview all domestic applicants and accept every eligible applicant. They also have to maintain a report that includes details of every applicant and contacted former worker, and an explanation of why any given applicant was not hired.

4. Once domestic recruitment has been completed and housing has been inspected by the SWA, the farmer can then apply for final determination of their labor certification with the OFLC

5. After the farmer obtains a labor certification, they can then submit a petition to the United States Citizenship and Immigration Services (USCIS) informing them of the guest workers the farmer intends to hire.

6. Once approved by USCIS, these guest workers will travel to their U.S. consulates to apply for a visa.

7. Once the workers obtain a visa, they can then travel to the U.S. to reach their worksite.

2.3 The Adverse Effect Wage Rate

The Adverse Effect Wage Rate (AEWR) is set by the Department of Labor, and serves as a wage floor for workers hired via the H-2A program. The AEWR is intended to prevent the hiring of H-2A workers from depressing the wages of domestic farmworkers. Farmers have to include information on the wages they intend to pay the workers they hire via the H-2A program, and this wage rate has to equal or exceed the AEWR, the agreed-upon collective bargaining wage, or the federal or state statutory minimum wage, whichever is highest. In practice, the AEWR is typically the relevant wage floor due to the way it is calculated. The DOL calculates the AEWR for different regions in the U.S. based primarily on wage data from the USDA's Farm Labor Survey (FLS), and, for occupations or regions not adequately covered by the FLS, the Bureau of Labor Statistics' Occupational Employment and Wage Statistics (OEWS). These rates are updated and published annually in the Federal Register and on the Department of Labor's website, near the beginning of the year. The AEWR is calculated in three different ways depending on the occupation; we summarize the DOL's methodology (United States Department of Labor 2025a).

For non-range workers that work in field or livestock occupations, the AEWR is set as an hourly rate, and is calculated from the average wage rates reported for all such such occupations, combined, in the FLS. These occupations are listed by their SOC codes,

- 45-2041 - Graders and Sorters, Agricultural Products.
- 45-2091 - Agricultural Equipment Operators.
- 45-2092 - Farmworkers and Laborers, Crop, Nursery, and Greenhouse.
- 45-2093 - Farmworkers, Farm, Ranch, and Aquacultural Animals.
- 53-7064 - Packers and Packagers, Hand.
- 45-2099 - Agricultural Workers, All Other.

, and they comprise the overwhelming majority of H-2A applications. The AEWR that applies to these occupations is the relevant wage rate for our research. Henceforth, discussion of the AEWR will refer to this rate unless otherwise specified. If any given region or state does not have coverage by the FLS for these occupations, the average wage rates from the state or national OEWS for these occupations, combined, is used instead. In practice, this only applies to the District of Columbia, Alaska, and U.S. Territories. The AEWR is set at a regional level, so its change relative to local wages can vary substantially within a region. Figure 2 maps the change in the commuting-zone wage-premium measure used below.

For all other non-range occupations not already covered, e.g., logging, trucking, and supervisors, the AEWR is set at the state and occupation level. This AEWR in any given state-occupation is simply the mean hourly wage as reported in the OEWS report for that state and SOC code. If the state OEWS is not available, the national OEWS is used instead. In the event that a worker's tasks encompass multiple SOCs, the highest wage rate of those SOCs apply.

For range and animal herding occupations, the AEWR is calculated as a monthly salary, and applies nationally to all workers in this category. Each year's AEWR is the previous year's AEWR inflated by the Bureau of Labor Statistics' Employment Cost Index. The Employment Cost Index is an analogue of the the Producer Price Index or Consumer Price Index, but applied to a fixed "basket" of labor, and measures the hourly cost of hiring labor for US employers, inclusive of benefits. This inflated AEWR is then carried forward into the calculation for next year's AEWR for range and animal herding occupations. This category of occupations was quite recently added to the H-2A program in 2016, and for that year the monthly AEWR was simply defined as $7.25 (the federal minimum hourly wage), multiplied by 48 hours, and then multiplied

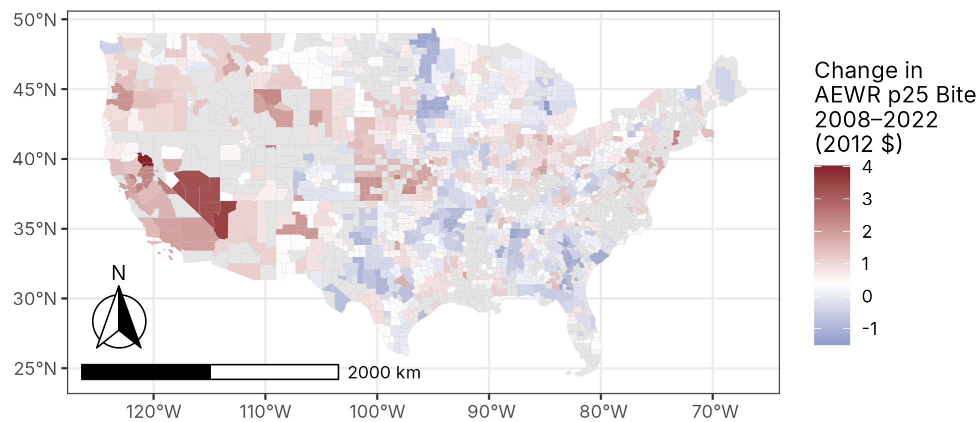

Figure 2: Change in the AEWR premium relative to the commuting-zone 25th-percentile wage, 2008–2022.

by 4.333 weeks per month. This number is then multiplied by 0.8 for 2016, and multiplied by 1 for 2017, and inflated using the index thereafter.

2.4 Additional costs of using the program

Farmers who use the program must pay for the travel costs incurred by the guest workers during this application process, provide or pay for lodgings, provide transportation for their workers' commute to their worksite and for weekly grocery runs, and all associated application fees. This alone incurs anywhere between \$10,000 to \$14,000 + weekly travel costs per worker hired using the H-2A program (United States Department of Agriculture 2025b).

An additional complication is that employers of H-2A workers have to pay all workers in "corresponding employment" at least the hourly rate they pay their H-2A workers plus an adjustment for the additional costs incurred by H-2A workers, i.e., lodgings and transport.

2.5 Changes in AEWR determination methodology

- Pre-1987:
  - History detailed in 54 FR 28037
  - Used 1950 Census of Agriculture state numbers and adjusted that upwards occasionally
- 1987-2008:
  - 54 FR 28037 final rule: establishes new methodology: set USDA FLS average hourly wage within each region in the previous year as the AEWR
  - Reasoning:
    * Previous methodology led to AEWR that was above prevailing agricultural wages in some states and below in others
- Feb 13, 2008:
  - Notice of Proposed Rulemaking on February 13, 2008 (73 FR 8538)
  - Proposes change from FLS to OES
  - Reasoning:
    * https://www.federalregister.gov/d/E8-29309/p-611
    * AEWR is too high, leading to farmers not utilizing the program and choosing to hire undocumented workers instead, potentially leading to wage depression
    * FLS inaccurate as it does not ask for hourly wage, and combines many occupations of different skill levels
    * FLS not administed by DOL, and is discretionary so USDA can terminate it at any time (it has almost done so many times, and actually did terminate it in August 2025)
    * OES is calculated at finer occupational and geographic granularity, so higher accuracy of estimated prevailing wage

    * More predictable budgeting by fixing contract wage at time contract signed, rather than adjusting with time work performed as under AEWR rule

- Jan 17, 2009
  - 73 FR 77110 Final Rule implementing switch to OES goes into effect
  - DOL no longer required to publish AEWR
  - Table 1 in 75 FR 6884 gives us the AEWRs in effect, calculated from OES numbers and provided to applicants
- May 29, 2009
  - 74 FR 25972 Final Rule implements suspension of 2008 Final Rule as previously proposed (74 FR 11408 on Mar 17 2009, 74 FR 17597 on Apr 16 2009)
- June 29, 2009
  - Injunction granted against suspension of 2008 Final Rule
  - Lawsuit brought by NCGA (NCGA v. Solis, 1:09-cv-00411)
- Sep 4, 2009
  - 74 FR 45906 Proposed rule for reverting back to FLS methodology
- Mar 15, 2010:
  - 75 FR 6884 Final Rule effective March 15, 2010
  - Return to FLS methodology
  - Reasoning:
    * OES does not survey farms, rejects assumption that farming support services capture farm wages
    * Rejects existence of differential skill level in agricultural labor, hence no need for OES
    * AEWR being too high for many locations due to wide geographic aggregation is feature, not bug (prevents localized wage depression due to program usage)

* Higher estimation errors for OES compared to FLS
* Changes in wages that occur between contract formation and performance are unavoidable and must be incorporated into employers' planning.

3 Data sources

3.1 H-2A program usage

The DOL's Office of Foreign Labor Certification (OFLC) posts disclosure files on H-2A program performance data on their website every year. The primary data source consists of H-2A disclosure files from fiscal years 2008 through 2022. These files contain detailed information on every H-2A application received in that year, and include the employer's name and address, the number of workers requested, the nature of the job, worksite addresses, job start and end dates, negotiated hourly wage rates, weekly work hours, and sometimes the relevant crops. The biggest challenge here is in assigning counties to worksites correctly. This is because, prior to 2017, the OFLC only had worksite location data by city, ZIP code, and state. The OFLC only included county information beginning in 2017. This is not even considering errors in data entry or transcription. We use a multi-tiered approach for matching entries that cannot be directly matched on county names. In order of matching priority, from highest to lowest:

1. Direct matching with ZIP codes: The vast majority of entries include ZIP code information. The U.S. Census Bureau provides crosswalks from ZIP codes to counties. Some ZIP codes map to multiple counties; in those cases we treat the entries as containing multiple counties. We describe below how entries with multiple counties are handled.

2. Direct Matching with place names: For entries with reported county names but no ZIP codes, counties can be matched directly. For entries with reported city names but no reported county (mostly entries prior to 2017), counties can be assigned using U.S. Census Bureau files that provide names for all incorporated places within every county. These places would include cities, town, boroughs, and villages. A modified procedure was implemented for New England states, where "towns", equivalent to cities and towns in other states, often serve as the primary unit of local government instead of counties, and thus county information is omitted. In these cases, the worksite "county" entries, which frequently contain town names, were matched with Census place names.

3. Fuzzy String Matching: For worksites that could not be directly matched using either place or county

names, a fuzzy string matching approach was used. When county names are available, we use fuzzy string matching with county names in the Census county file and select the best match. This takes care of typographical errors in the reported county names. For entries without county names, we calculate string-similarity scores between worksite city names reported in the H-2A data and all Census place names within the same state. Algorithms and score cutoffs were selected case by case to minimize false positives while capturing legitimate name variations.

4. Google Places API: For the remaining unmatched locations, we use Google's Places API as a geocoding solution. These are mostly worksites that are located in unincorporated areas, or reported wity unusual place names, e.g., landmarks, crossroads, etc. This involved a two-step process. First, the "Find Place" endpoint was queried with the reported worksite city and state to obtain a unique location ID. Second, the "Place Details" endpoint was queried with the location ID to retrieve the full address components, including the county name.

The cleaned and geocoded entries were then aggregated to the county-year level. For entries with multiple counties, either because the matched location spans multiple counties or because the employer reports multiple worksites for the same workers, we split man-hours and workers evenly across counties in the entry. To account for employment periods spanning across calendar years, the number of workers and man-hours for these applications were weighted by the proportion of the contract period falling within each calendar year.

The final dataset includes the following aggregated variables for each county-year:

- Total number of H-2A workers requested
- Total number of H-2A workers certified
- Total requested man-hours
- Total certified man-hours
- Total number of applications

3.2 Adverse Effect Wage Rates (AEWR)

We collect historical AEWR values from annual notices published the Department of Labor in the Federal Register. We identify Department of Labor notices in the Federal Register containing the keywords "labor certification process" and "adverse effect wage rates" and download the relevant pages. Because the annual notices use a consistent table format, the extraction is automated to construct an AEWR panel by state and year.

3.3 Crop acreage

To measure land use patterns, we draw on two USDA National Agricultural Statistics Service (NASS) sources. The first is the Census of Agriculture, conducted every five years (in years ending in ’2’ and ’7’). It provides county-level data on the acreage of hundreds of specific crops. The second is the Cropland Collaborative Research Outcomes System (CroplandCROS) project. This project classifies 30-by-30-meter parcels of land in the U.S. by use and crop type and publishes the data in its Cropland Data Layer (CDL) product. Each parcel of land corresponds to a single pixel in the CDL. We assign each pixel to a county and aggregate pixels by county-crop-year.

3.4 Crop prices and yields

We obtain state-level crop prices and yields from agricultural surveys that the NASS conducts every year. While county-level prices and yields would be ideal, they are not systematically available for specialty crops. We also have county-level prices from USDA’s Risk Management Agency actuarial data, however this obviously only ccovers specific county-crop-years for which federal crop insurance policies were purchased.

3.5 Occupational Employment and Wage Statistics (OEWS)

The Occupational Employment and Wage Statistics (OEWS) program of the BLS produces estimates of employment and wages for about 830 occupations across the U.S., by industry and locality. The program surveys every non-farm establishment registered with State Workforce Agencies (SWAs), which excludes most of the agricultural sector. However, the survey does include establishments that fall under NAICS 1151 - Support Activities for Crop Production, which includes establishments engaged in farm labor contracting. Note that the DOL uses OEWS estimates to calculate the AEWR for localities and occupations not adequately covered by the FLS. The estimates are available at the OEWS Metropolitan and Nonmetropolitan Area level, and geographic crosswalks permit county-level estimates. We calculate the county estimate as the employment-weighted mean of the OEWS locality estimates available within the county. The occupations used are the same occupations included in the FLS, with SOC codes

- 45-2041 - Graders and Sorters, Agricultural Products.
- 45-2091 - Agricultural Equipment Operators.

- 45-2092 - Farmworkers and Laborers, Crop, Nursery, and Greenhouse.
- 45-2093 - Farmworkers, Farm, Ranch, and Aquacultural Animals.
- 53-7064 - Packers and Packagers, Hand.
- 45-2099 - Agricultural Workers, All Other.

We can also construct an AEWR wage equivalent by calculating the mean wage for these occupations, which is the same procedure used by the DOL.

3.6 Sample construction

The analysis panel covers 2008–2022. We retain counties with positive cropland in 2007 and exclude counties classified as "always takers" by the project's treatment-group construction. Alaska, Hawaii, the District of Columbia, and observations without an AEWR-region assignment are not in the analysis file. The baseline panel contains 145,000 county-year observations; the small difference from the balanced count reflects data availability for the outcome and controls. Table 1 reports summary statistics.

Table 1: Summary Statistics: Difference-in-Differences Variables

Variable	N	Mean	SD	Min	Max
H-2A share of 2011 farm employment	145,012	0.036	0.327	0.000	20.569
H-2A certified workers (start year)	145,012	23.791	139.512	0.000	5263.721
Farm employment 2011 (baseline)	145,012	855.841	1049.345	12.000	20090.000
AEWR p25 bite (2012 $)	145,012	1.588	1.498	-7.897	7.164
Log population	145,012	10.188	1.411	5.451	16.129
Employment-to-population ratio	145,012	0.512	0.147	0.107	2.155

4 Empirical design and results

4.1 Design

Let Y_{ct} be certified H-2A workers with a start date in year t, divided by farm employment in county c in 2011. Define the lagged wage-premium measure as

$$B_{ct-1} = \mathrm{AEWR}^{\mathrm{real}}_{ct-1} - w^{p25,\mathrm{real}}_{\mathrm{CZ}(c),t-1},$$

where the wage percentile is measured within the county's commuting zone and all wage values are expressed in 2012 dollars. The baseline regression is

$$Y_{ct} = \beta B_{ct-1} + \delta \left(B_{ct-1} \times \mathbb{1}\{t > 2011\} \right) + X_{ct}'\gamma + \alpha_c + \tau_t + \varepsilon_{ct}.$$

County fixed effects (α_c) absorb time-invariant differences in local agricultural production and H-2A use, while year fixed effects (τ_t) absorb common shocks. The controlled specification includes log population and the employment-to-population ratio. Standard errors are clustered by the commuting-zone-by-AEWR-region cell. We report a complementary sample that excludes commuting zones crossing AEWR-region borders, where the interpretation of a commuting-zone wage comparison is less direct.

The event-study specification replaces the single post-2011 interaction with interactions between B_{ct-1} and each calendar year, omitting 2011. It therefore tests whether high- and low-premium counties had different trends before the post period and traces the evolution of the estimated relationship.

4.2 H-2A utilization

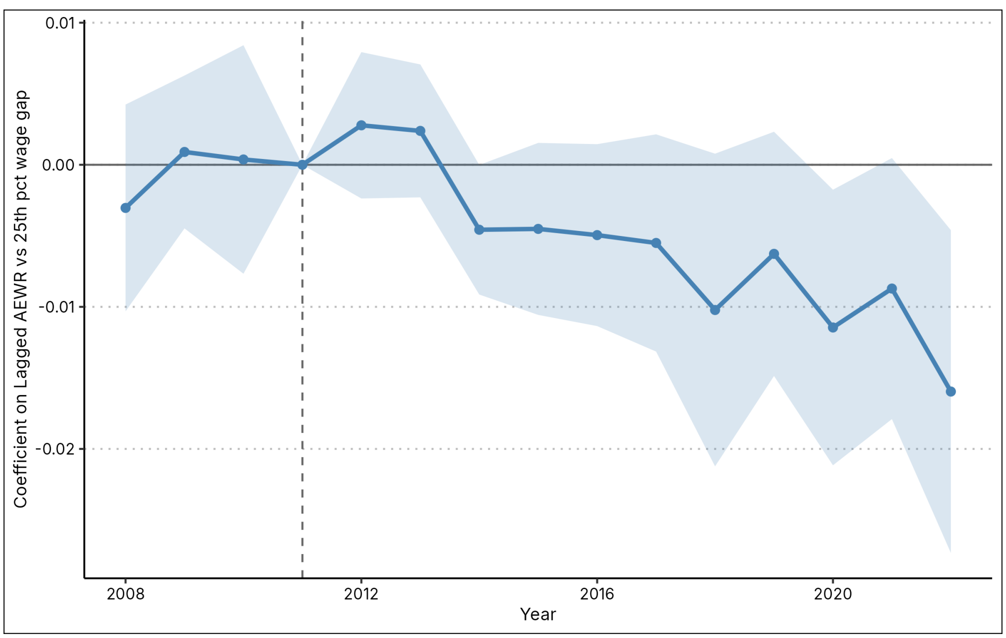

Figure 3: Event-study coefficients for the lagged AEWR wage premium, with controls. The omitted year is 2011; shaded bands are 95% confidence intervals.

Table 2 reports the main estimates. In column (2), the coefficient on the post-2011 interaction is -0.0061 (standard error 0.0035), and in the no-border controlled sample it is -0.0075 (standard error 0.0044). Thus, the negative relationship is not driven by the commuting zones that cross AEWR-region borders. These

Dependent Variable:	Normalized H-2A program usage			
	No Controls	Controls	No Border, No Controls	No Border, Controls
Model:	(1)	(2)	(3)	(4)
Variables				
Lagged AEWR vs 25th pct wage gap	0.0036	0.0030	0.0039	0.0033
	(0.0036)	(0.0035)	(0.0044)	(0.0043)
Lagged AEWR vs 25th pct wage gap × Post	-0.0082**	-0.0061*	-0.0096**	-0.0075*
	(0.0037)	(0.0035)	(0.0047)	(0.0044)
Log population		0.1187		0.1189
		(0.1135)		(0.1413)
Employment-to-pop ratio		0.2406***		0.2481**
		(0.0920)		(0.1031)
Fixed-effects				
county_fe	Yes	Yes	Yes	Yes
year_fe	Yes	Yes	Yes	Yes
Fit statistics				
Observations	145,000	145,000	119,016	119,016
R^2	0.43468	0.43526	0.43511	0.43561
Within R^2	0.00047	0.00149	0.00052	0.00140

Clustered (cz_aewr_region_fe) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 2: Difference-in-differences estimates of the relationship between the lagged AEWR wage premium and normalized H-2A utilization.

magnitudes are percentage points of 2011 farm employment, not percent changes in the number of certifications.

Figure 3 and Table 3 show the flexible event-study results. The coefficients in 2008–2010 are small relative to their standard errors. The post-period estimates turn negative in 2014 and are largest in 2022, when the controlled estimate is -0.0160 (standard error 0.0058). The pattern is consistent with a growing negative association between the AEWR premium and H-2A use, while the limited number of AEWR regions and the observational nature of the wage premium warrant cautious interpretation.

4.3 Robustness to exposure definitions

The current specification uses a level difference between the AEWR and the commuting-zone 25th-percentile wage. Table 4 compares this measure with a positive log premium and with a specification that absorbs AEWR-region-by-year fixed effects. The positive-log-premium estimate remains negative and statistically distinguishable from zero. By contrast, the region-year fixed-effect specification is imprecise. This comparison shows that the baseline result is informative about variation in the regional wage premium over time, but it

Dependent Variable:	h2a_cert_share_farm_workers_2011_start_year			
	No Controls	Controls	No Border, No Controls	No Border, Controls
Model:	(1)	(2)	(3)	(4)
Variables				
Lagged AEWR vs 25th pct wage gap × 2008	-0.0034	-0.0030	-0.0047	-0.0043
	(0.0037)	(0.0037)	(0.0046)	(0.0046)
Lagged AEWR vs 25th pct wage gap × 2009	0.0014	0.0009	0.0012	0.0007
	(0.0028)	(0.0027)	(0.0034)	(0.0034)
Lagged AEWR vs 25th pct wage gap × 2010	-0.0002	0.0004	-6.56×10^{-5}	0.0005
	(0.0043)	(0.0041)	(0.0053)	(0.0050)
Lagged AEWR vs 25th pct wage gap × 2012	0.0030	0.0028	0.0030	0.0029
	(0.0027)	(0.0026)	(0.0033)	(0.0032)
Lagged AEWR vs 25th pct wage gap × 2013	0.0024	0.0024	0.0026	0.0025
	(0.0026)	(0.0024)	(0.0031)	(0.0029)
Lagged AEWR vs 25th pct wage gap × 2014	-0.0052^{**}	-0.0046^{**}	-0.0068^{**}	-0.0062^{**}
	(0.0025)	(0.0023)	(0.0032)	(0.0029)
Lagged AEWR vs 25th pct wage gap × 2015	-0.0054	-0.0045	-0.0063	-0.0054
	(0.0037)	(0.0031)	(0.0045)	(0.0038)
Lagged AEWR vs 25th pct wage gap × 2016	-0.0063^{*}	-0.0050	-0.0075	-0.0062
	(0.0037)	(0.0033)	(0.0047)	(0.0041)
Lagged AEWR vs 25th pct wage gap × 2017	-0.0075^{*}	-0.0055	-0.0095^{*}	-0.0075
	(0.0045)	(0.0039)	(0.0056)	(0.0048)
Lagged AEWR vs 25th pct wage gap × 2018	-0.0124^{**}	-0.0102^{*}	-0.0154^{**}	-0.0133^{*}
	(0.0060)	(0.0056)	(0.0076)	(0.0071)
Lagged AEWR vs 25th pct wage gap × 2019	-0.0088^{*}	-0.0063	-0.0110^{*}	-0.0086
	(0.0050)	(0.0044)	(0.0063)	(0.0055)
Lagged AEWR vs 25th pct wage gap × 2020	-0.0139^{**}	-0.0115^{**}	-0.0157^{**}	-0.0134^{**}
	(0.0054)	(0.0049)	(0.0067)	(0.0061)
Lagged AEWR vs 25th pct wage gap × 2021	-0.0115^{**}	-0.0087^{*}	-0.0130^{*}	-0.0103^{*}
	(0.0056)	(0.0047)	(0.0069)	(0.0057)
Lagged AEWR vs 25th pct wage gap × 2022	-0.0194^{***}	-0.0160^{***}	-0.0234^{***}	-0.0201^{***}
	(0.0060)	(0.0058)	(0.0074)	(0.0073)
Fixed-effects				
county_fe	Yes	Yes	Yes	Yes
year_fe	Yes	Yes	Yes	Yes
Fit statistics				
Observations	145,000	145,000	119,016	119,016
R^2	0.43509	0.43554	0.43557	0.43594
Within R^2	0.00118	0.00198	0.00132	0.00199

Clustered (cz_aewr_region_fe) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 3: Event-study estimates; 2011 is the omitted year.

does not separately identify within-region changes once every AEWR-region-year shock is absorbed.

Dependent Variable:	h2a_cert_share_farm_workers_2011_start_year			
	Level Bite	Level Bite + Controls	Positive Log Bite	Level Bite + AEWR Re
Model:	(1)	(2)	(3)	(4)
Variables				
Lagged AEWR - CZ p25 wage	0.0036	0.0030		-0.0016
	(0.0036)	(0.0035)		(0.0049)
Lagged AEWR - CZ p25 wage × Post-2011	-0.0082**	-0.0061*		0.0131
	(0.0037)	(0.0035)		(0.0080)
Log population		0.1187		
		(0.1135)		
Employment-to-pop ratio		0.2406***		
		(0.0920)		
Positive log AEWR/CZ p25 wage			0.1175	
			(0.0714)	
Positive log AEWR/CZ p25 wage × Post-2011			-0.1250***	
			(0.0442)	
Fixed-effects				
county_fe	Yes	Yes	Yes	Yes
year_fe	Yes	Yes	Yes	
aewrregtime_fe				Yes
Fit statistics				
Observations	145,000	145,000	145,000	145,000
R^2	0.43468	0.43526	0.43468	0.45379
Within R^2	0.00047	0.00149	0.00047	0.00077

Clustered (cz_aewr_region_fe) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 4: Sensitivity of the main difference-in-differences estimate to alternative wage-premium definitions.

4.4 Exploratory downstream outcomes

The code also constructs a county Fisher agricultural price index, chained from crop-level prices and acreage and expressed in real 2012 dollars. In the same reduced-form specification, the post-2011 premium interaction is negative in all four price regressions (Table 5). The controlled baseline estimate is -1.439 index points per dollar (standard error 0.629). This result is exploratory: it combines price and crop-composition changes and does not identify a pass-through mechanism.

The corresponding estimates for the labor share of farm production expenses are small and statistically imprecise after 2011 (Table 6). Accordingly, the current evidence does not establish that the H-2A response operates through a shift in the labor share of farm production expenses.

Dependent Variable:	Fisher price index (real, 2012=100)			
	No Controls	Controls	No Border, No Controls	No Border, Controls
Model:	(1)	(2)	(3)	(4)
Variables				
Lagged AEWR vs 25th pct wage gap	0.4655	0.3211	0.6805	0.5109
	(0.3995)	(0.3796)	(0.4789)	(0.4525)
Lagged AEWR vs 25th pct wage gap × Post	-1.940***	-1.439**	-2.243***	-1.626**
	(0.6970)	(0.6292)	(0.8490)	(0.7629)
Log population		31.87***		39.56***
		(9.188)		(10.76)
Employment-to-pop ratio		6.280		10.22
		(5.869)		(6.586)
Fixed-effects				
county_fe	Yes	Yes	Yes	Yes
year_fe	Yes	Yes	Yes	Yes
Fit statistics				
Observations	144,792	144,792	118,860	118,860
R^2	0.55673	0.56000	0.54497	0.54942
Within R^2	0.00667	0.01399	0.00747	0.01718

Clustered (cz_aewr_region_fe) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 5: Exploratory reduced-form estimates for the county Fisher agricultural price index.

4.5 Sampling-composition instrument: current status

The project contains an extension that seeks to isolate variation in the AEWR arising from the geographic composition of the USDA Farm Labor Survey sample. It starts from BEA farm-employment weights, uses entropy calibration to describe deviations from those expected weights, and constructs a leave-subregion-out donor wage from OEWS agricultural wages. The implemented code currently reports first-stage diagnostics for changes in the AEWR. For example, the 50% gap-closure version has a clustered first-stage F statistic of about 103 for nominal AEWR changes with commuting-zone-by-region and year fixed effects. This strength is useful for development, but it is not an IV estimate and must not be interpreted as one. In particular, calibrating weights directly to the FLS wage that sets the AEWR would create a mechanical first stage. The preferred instrument will require auxiliary FLS moments that exclude that wage-setting moment.

5 Model

- Production:
  - Mass of farms, indexed by $i \in [0, 1]$

Dependent Variable:	Labor share of farm production expense			
	No Controls	Controls	No Border, No Controls	No Border, Controls
Model:	(1)	(2)	(3)	(4)
Variables				
Lagged AEWR vs 25th pct wage gap	0.0010***	0.0010***	0.0012***	0.0011***
	(0.0003)	(0.0003)	(0.0004)	(0.0004)
Lagged AEWR vs 25th pct wage gap × Post	-0.0003	-0.0001	-0.0005	-0.0004
	(0.0004)	(0.0004)	(0.0005)	(0.0005)
Log population		0.0116*		0.0114
		(0.0062)		(0.0074)
Employment-to-pop ratio		0.0029		0.0014
		(0.0083)		(0.0093)
Fixed-effects				
county_fe	Yes	Yes	Yes	Yes
year_fe	Yes	Yes	Yes	Yes
Fit statistics				
Observations	145,000	145,000	119,016	119,016
R^2	0.94133	0.94138	0.94485	0.94490
Within R^2	0.00131	0.00220	0.00168	0.00251

Clustered (cz_aewr_region_fe) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 6: Exploratory reduced-form estimates for labor's share of farm production expenses.

  - Farms are heterogeneous in farm size/productivity/crop suitability

$$A_i \sim G(A_i)$$

  - Farms use labor L to produce output with production function

$$Y_i = A_i f(L_i)$$

    where $f'(L) > 0$ and $f''(L) < 0$

  - Farms receive price p for their output

- Labor market:

  - Farms have 2 sources of labor, Domestic and H-2A.

  - Domestic labor: hire $L_{i,D}$, pay prevailing Domestic wage w_D

- Domestic labor market is tight:

$$\int_0^1 L_{i,D} d_i = \bar{L}$$

- H-2A labor: hire $L_{i,F}$, pay AEWR w_F to ALL workers, including domestic
- H-2A labor market is slack, no constraints

5.1 Farmer's Problem

- Use only Domestic labor, profits are

$$\Pi_D(A_i) = pAf(L_{i,D}) - w_D L_{i,D}$$

- Use H-2A labor, profits are

$$\Pi_F(A_i) = pA_i f(L_{i,F} + L_{i,D}) - w_F(L_{i,F} + L_{i,D}) - F$$

  where F is the fixed cost of participating in the H-2A program

- H-2A adoption decision: compare $\Pi_D(A_i)$ and $\Pi_F(A_i)$
- Simplification: only 1 farm, so $L_{i,D} = \bar{L}$
- Then

$$\Pi_D(A_i) = pAf(\bar{L}) - w_D \bar{L}$$

$$\Pi_F(A_i) = pA_i f(L_{i,F} + \bar{L}) - w_F(L_{i,F} + \bar{L}) - F$$

- Denote total labor hired as $L_i^* = L_{i,F} + \bar{L}$. Then

$$\Pi_F(A_i) - \Pi_D(A_i) = \underbrace{\int_{\bar{L}}^{Li^*} (pAf'(L) - w_F)\, dL}_{\text{Expansion gain}} - \underbrace{(w_F - w_D)\bar{L}}_{\text{Corresponding employment wage penalty on Domestic workers}} - F$$

- Use H-2A program if profit gain from expansion outweighs the wage penalty paid to Domestic workers + fixed costs
- Comparative statics:

- Counties with lower AEWR wage growth relative to local wages ($w_F - w_D$) $\downarrow$ see higher H-2A adoption
- Counties with larger pre-existing domestic farm labor (big $\bar{L}$) should see less adoption

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